

Triangular Probability Distribution

In this lab, we will become familiar with the Triangular probability distribution. In particular,

1. We will use the PDF of this distribution to plot it.
2. We will use the CPF of this distribution to calculate probabilities.
3. We will generate random values according to a specific Triangular distribution.
4. We will perform the Chi-squared Goodness-of-fit test to check how well the Triangular distribution fits our simulated values.

Theoretical Backgrounds:

The triangular probability distribution (also called: “a lack of knowledge distribution”) is a simplistic continuous model that is mainly used in situations when there is only limited sample data and information about a population. It is based on the knowledge of a minimum (a lower value), a maximum (an upper value), and a mode (peak) between those two values. For this reason, this distribution is very popular in simulation processes related to business decision models, project management models, financial models, and for modeling noises in digital audio and video data.

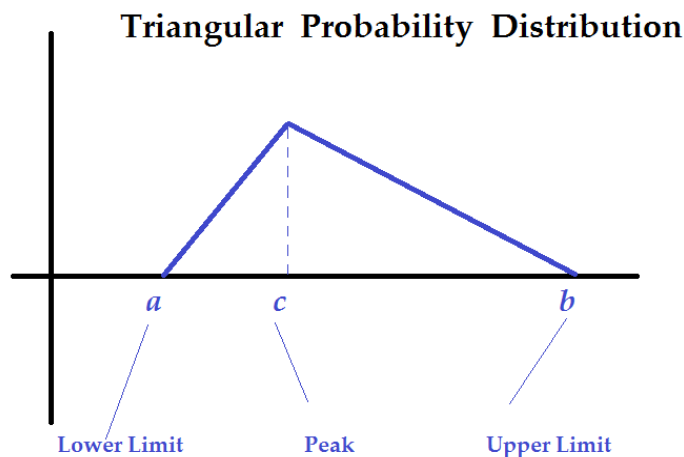


Figure 1: A Triangular Probability Distribution

The probability density function (*pdf*) of the triangular random variable X is given by the following expression. Note that the pdf of a continuous distribution is mainly used for plotting the graph of that distribution.

$$f(x) = \begin{cases} \frac{2}{(b-a)(c-a)} (x-a) & \text{if } x \leq c \\ \frac{2}{(b-a)(b-c)} (b-x) & \text{if } x > c \end{cases} \quad (1)$$

The following are some of the important numerical characteristics of the triangular distribution:

$$\text{mean} = E(x) = \mu = \frac{a+b+c}{3} \quad (2)$$

$$\text{variance} = \sigma^2 = \frac{a^2+b^2+c^2-ab-ac-bc}{18} \quad (3)$$

The *cdf* (cumulative density function) $P(X \leq x)$ of the triangular random variable is:

$$F(x) = \begin{cases} \frac{1}{(b-a)(c-a)} (x-a)^2 & \text{if } x < c \\ \frac{c-a}{b-a} & \text{if } x = c \\ 1 - \frac{1}{(b-a)(b-c)} (b-x)^2 & \text{if } x > c \end{cases} \quad (4)$$

For example, the following is a display of the cumulative density function of a triangular random variable with a minimum value of 2, a maximum of 8, and with a peak at 4.

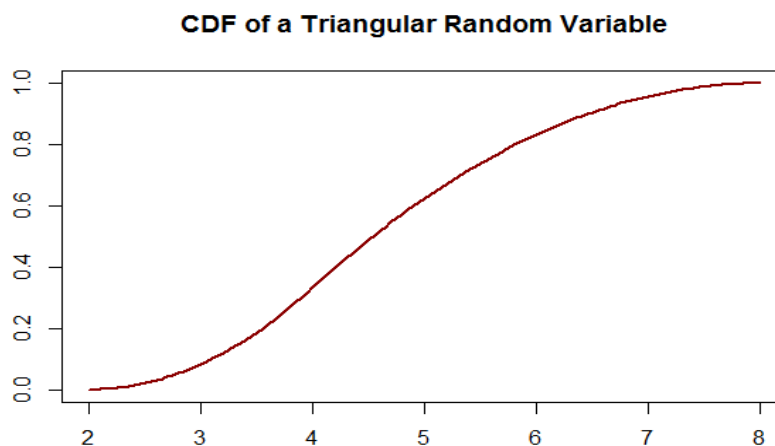


Figure 2: The CDF of a Triangular Probability Distribution

Random Number Generation of Triangular Random Variables:

The CDF expression given by formula (4) can be used to generate random values according to a specific triangular distribution. In this method, first a standard uniform random value r is created. This value is then used as a cumulative probability and replaces $F(x)$ in formula (5). The formula is then solved for the random variable x . The following rule describes this random number generation:

$$x = \begin{cases} a + \sqrt{r(b-a)(c-a)} & \text{if } r \leq \frac{c-a}{b-a} \\ b - \sqrt{(1-r)(b-a)(b-c)} & \text{if } r > \frac{c-a}{b-a} \end{cases} \quad (5)$$